Why does the magnetic-force expression involve a cross product?

Electrostatics versus current elements. In electrostatics, the force between two point charges lies along the joining vector. For two current elements, however, experiments show that the force direction is not so simple.

$$\mathbf{r}_{12} = \mathbf{r}_2 - \mathbf{r}_1$$

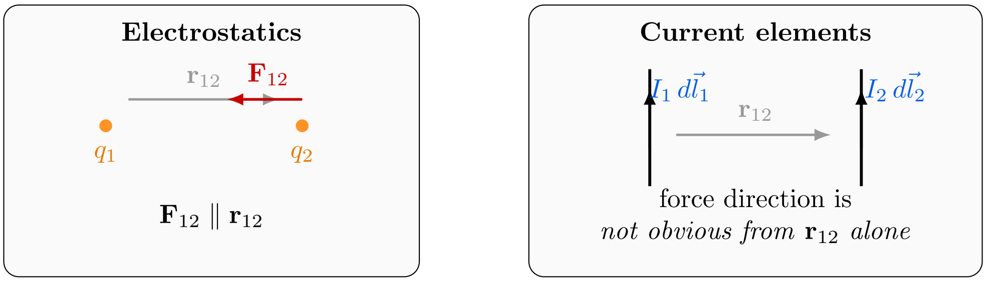


Figure 1: Comparison: Coulomb force acts along the joining vector, whereas the magnetic interaction between current elements has a different directional structure.

A natural question. Can we directly guess a force law for two current elements from $I_1 d\vec{l}_1$, $I_2 d\vec{l}_2$, and $\vec{r}_{12}$? For example, one may try vector expressions such as

$$\left(I_1 d\vec{l}_1 \cdot I_2 d\vec{l}_2\right) \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3}, \qquad \left(I_1 d\vec{l}_1 \cdot \frac{\vec{r}_{12}}{|\vec{r}_{12}|^3}\right) I_2 d\vec{l}_2.$$

But experiments show that such simple guesses do not correctly capture the direction of force in all cases.

Key experimental observations

1. When the current element and the separation vector are collinear, the force is observed to be zero.
2. In the perpendicular arrangement sketched below, the force is also zero.

These observations strongly suggest that the magnetic interaction must involve a **cross product**.

Observation 1: collinear arrangement gives zero force.

Observation 2: a perpendicular arrangement can also give zero force.

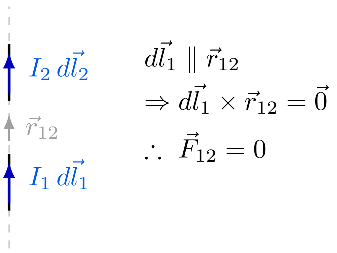


Figure 2: If the source current element and the separation vector are collinear, the force is zero.

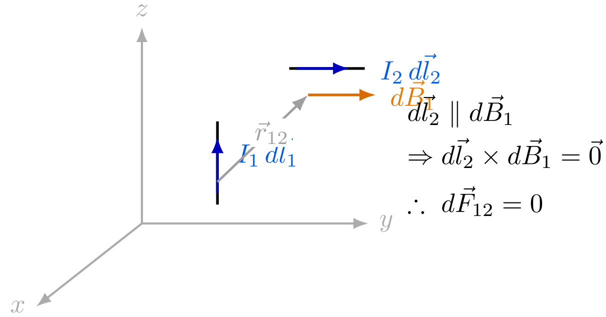


Figure 3: The vector $\vec{r}_{12}$ is drawn from the source element $I_1\,d\vec{l}_1$ to the location of the second element (field point). If $d\vec{l}_2$ is parallel to the local magnetic-field direction $d\vec{B}_1$, then the force is zero.

This suggests the correct vector structure. The source element $I_1 d\vec{l}_1$ must first create a field whose direction is perpendicular to both $d\vec{l}_1$ and $\vec{r}_{12}$. The simplest such vector is

$$d\vec{l}_1 \times \vec{r}_{12}.$$

This motivates the magnetic-field contribution of the first current element as

$$\boxed{d\vec{B}_1 = \frac{\mu_0}{4\pi}\frac{I_1\,d\vec{l}_1 \times \vec{r}_{12}}{|\vec{r}_{12}|^3}}$$

or equivalently,

$$d\vec{B}_1 = \frac{\mu_0}{4\pi}\frac{I_1\,d\vec{l}_1 \times \hat{\mathbf{r}}_{12}}{|\vec{r}_{12}|^2}.$$

Force on the second current element. Once the magnetic field due to element 1 is known, the force on the second current element is

$$\boxed{d\vec{F}_{12} = I_2\,d\vec{l}_2 \times d\vec{B}_1}$$

Therefore,

$$\boxed{d\vec{F}_{12} = \frac{\mu_0}{4\pi} I_2\,d\vec{l}_2 \times \left(\frac{I_1\,d\vec{l}_1 \times \vec{r}_{12}}{|\vec{r}_{12}|^3}\right)}$$

which is the differential magnetic-force expression suggested by these experimental observations.

Interpretation

The double-cross-product form is not arbitrary. It is exactly what allows the force to satisfy the observed directional properties:

- if $d\vec{l_1} \parallel \vec{r}_{12}$, then $d\vec{B}_1 = \vec{0}$,
- if $d\vec{l_2} \parallel d\vec{B}_1$, then $d\vec{F}_{12} = \vec{0}$.

Key Takeaways

1. Current-carrying wire sections running in parallel can exert a significant force on each other.
2. In the idealized perpendicular arrangement sketched here, the magnetic force becomes zero.
3. The magnetic interaction is naturally described using two successive cross products:

$$d\vec{B}_1 \propto d\vec{l_1} \times \vec{r}_{12}, \qquad d\vec{F}_{12} \propto d\vec{l_2} \times d\vec{B}_1.$$

Biot–Savart law and the force on a current element

Analogy with electrostatics. In electrostatics, once the electric field is known, the force on a charge is

$$\boxed{\mathbf{F} = q\,\mathbf{E}}$$

and the field due to a point charge q' located at $\mathbf{r}'$ is

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\, q'\, \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}}.$$

Similarly, in magnetostatics, once the magnetic field is known, the force on a current element is

$$\boxed{d\mathbf{F} = I_2\, d\boldsymbol{\ell}_2 \times \mathbf{B}}$$

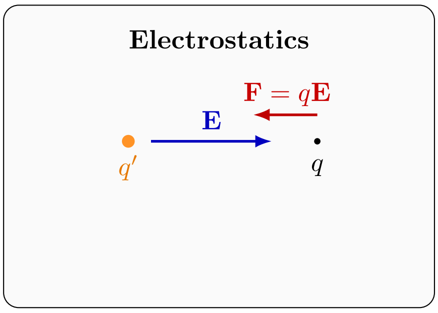
and the magnetic field contribution due to a source current element $I_1\, d\boldsymbol{\ell}_1$ is

$$\boxed{d\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi}\, I_1\, d\boldsymbol{\ell}_1 \times \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}}$$

which is the **Biot–Savart law** in differential form.

Important remark

In experiments, **force is what is directly measurable**. The magnetic field $\mathbf{B}$ is introduced as a very useful and compact mathematical description of how current elements interact.



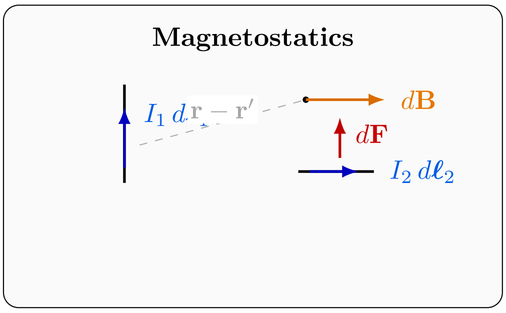


Figure 4: Analogy: in electrostatics, a charge experiences force in an electric field; in magnetostatics, a current element experiences force in a magnetic field.

Magnetic field due to a continuous current distribution

From many current elements to volume current density. If the current is distributed continuously in a volume, we replace the discrete current elements by the current density $\mathbf{J}$. For a small volume element,

$$I = J\, dA$$

and therefore

$$I\, d\boldsymbol{\ell} = (J\, dA)\, d\boldsymbol{\ell}.$$

Since

$$dV = dA\, d\ell,$$

we may write

$$\boxed{I\, d\boldsymbol{\ell} = \mathbf{J}\, dV}$$

when $d\boldsymbol{\ell}$ is taken along the direction of current flow.
Hence, summing all source contributions gives

$$\boxed{\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint_{\text{vol}} \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r} - \mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|^3}\, dV'}$$

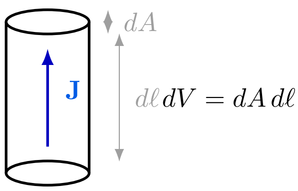


Figure 5: A small current-carrying volume element: $I = J\, dA$, and $I\, d\boldsymbol{\ell} = \mathbf{J}\, dV$.

Force on a current density

For a small current element inside a magnetic field,

$$d\mathbf{F} = I\, d\boldsymbol{\ell} \times \mathbf{B}.$$

Using

$$I\, d\boldsymbol{\ell} = \mathbf{J}\, dV,$$

we get

$$\boxed{d\mathbf{F} = (\mathbf{J} \times \mathbf{B})\, dV}$$

Therefore, the **force per unit volume** is

$$\boxed{\mathbf{f} = \mathbf{J} \times \mathbf{B}}$$

Key Takeaways

1. Biot–Savart law gives the magnetic-field contribution of a current element.
2. For a continuous current distribution, replace $I\, d\boldsymbol{\ell}$ by $\mathbf{J}\, dV$.
3. The magnetic force density is

$$\mathbf{f} = \mathbf{J} \times \mathbf{B}.$$

Comparison: electrostatics and magnetostatics

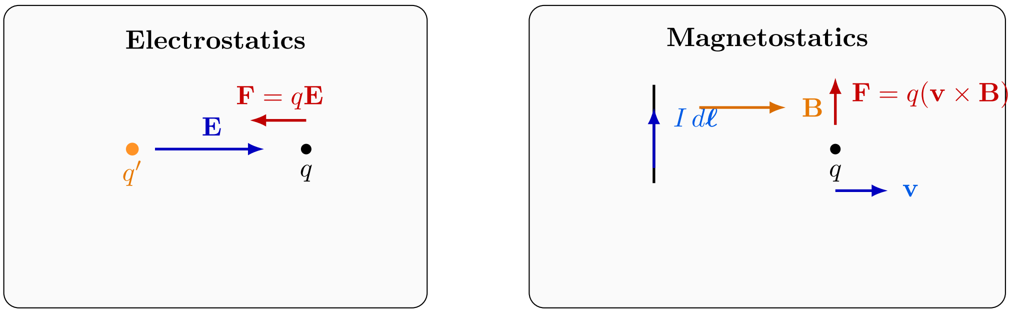


Figure 6: Basic comparison: charges produce electric fields, while currents produce magnetic fields; the corresponding force laws are different.

Electrostatics. For two point charges,

$$\boxed{\mathbf{F}_{12} = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{|\mathbf{r}_{12}|^3}\, \mathbf{r}_{12}}$$

and the electric field due to a continuous charge distribution is

$$\boxed{\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \iiint \rho(\mathbf{r}') \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}\, dV'}$$

Hence, the force on a charge q is

$$\boxed{\mathbf{F} = q\mathbf{E}.}$$

Magnetostatics. For current elements, the differential force may be written as

$$\boxed{d\mathbf{F}_{12} = \frac{\mu_0}{4\pi} I_2\, d\boldsymbol{\ell}_2 \times \left(I_1\, d\boldsymbol{\ell}_1 \times \frac{\mathbf{r}_{12}}{|\mathbf{r}_{12}|^3} \right)}$$

and the magnetic field due to a continuous current density is

$$\boxed{\mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \iiint \mathbf{J}(\mathbf{r}') \times \frac{\mathbf{r}-\mathbf{r}'}{|\mathbf{r}-\mathbf{r}'|^3}\, dV'}$$

The magnetic force on a moving charge is

$$\boxed{\mathbf{F} = q(\mathbf{v} \times \mathbf{B}).}$$

Total electromagnetic force (Lorentz force). If both electric and magnetic fields are present, then

$$\boxed{\mathbf{F} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right).}$$

Comparison in words

- **Electrostatics:** charges are the sources of **E**, and the force is along **E**.
- **Magnetostatics:** currents are the sources of **B**, and magnetic force acts only on a *moving* charge.
- Therefore, magnetism naturally appears together with motion.

Auxiliary fields in material media

Electrostatic side. For a linear isotropic medium,

$$\boxed{\mathbf{D} = \varepsilon\mathbf{E}}$$

and Gauss's law is written as

$$\boxed{\nabla \cdot \mathbf{D} = \rho_{\text{free}}.}$$

Also, in electrostatics,

$$\boxed{\mathbf{E} = -\nabla\phi.}$$

Magnetostatic side. For a linear isotropic medium,

$$\boxed{\mathbf{B} = \mu \mathbf{H}} \qquad \left(\text{equivalently } \mathbf{H} = \frac{\mathbf{B}}{\mu}\right)$$

and Ampère's law is written as

$$\boxed{\nabla \times \mathbf{H} = \mathbf{J}_{\text{free}}.}$$

Further, since

$$\boxed{\nabla \cdot \mathbf{B} = 0,}$$

we introduce the magnetic vector potential $\mathbf{A}$ such that

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.}$$

Important remark

The constitutive relations

$$\mathbf{D} = \varepsilon \mathbf{E}, \qquad \mathbf{B} = \mu \mathbf{H}$$

do depend on the medium. What is useful is that the field equations

$$\nabla \cdot \mathbf{D} = \rho_{\text{free}}, \qquad \nabla \times \mathbf{H} = \mathbf{J}_{\text{free}}$$

are written directly in terms of *free* charge and *free* current.

Key Takeaways

1. $\mathbf{E}$ is produced by charge, whereas $\mathbf{B}$ is produced by current.
2. Electric force acts on any charge, but magnetic force acts only on a moving charge.
3. The total force is the Lorentz force:

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}).$$

4. In materials, $\mathbf{D}$ and $\mathbf{H}$ are auxiliary fields used to handle free sources conveniently.

Magnetic field around a straight current-carrying wire

Consider a long straight wire carrying current I along $+\hat{z}$. From symmetry, the magnetic field circulates around the wire.
Thus, for a long straight wire,

$$\boxed{\mathbf{B} = B(r)\,\hat{\phi}}$$

that is,

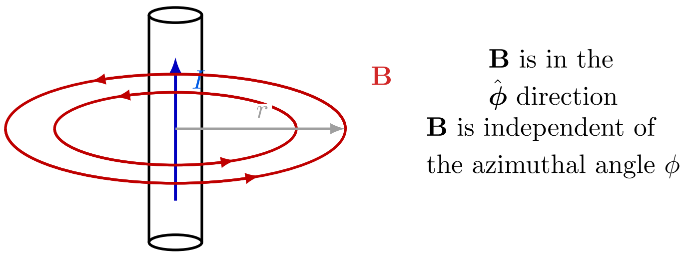


Figure 7: Magnetic field around a long straight wire: the field lines are circular, the direction is $\hat{\boldsymbol{\phi}}$, and the magnitude depends only on the radial distance r.

- **B** is along the azimuthal direction $\hat{\boldsymbol{\phi}}$,
- **B** is independent of ϕ,
- the magnitude varies only with the distance r from the wire.

Magnetic field lines do not begin or end at a point

In electrostatics, electric field lines can begin on positive charges and end on negative charges. Magnetic field lines, however, form closed loops. They do not converge to or diverge from an isolated point.

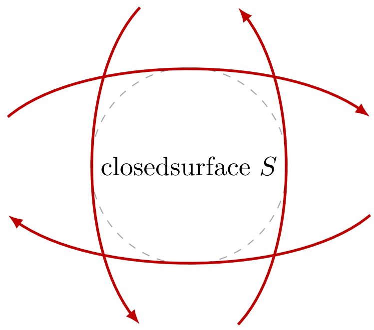


Figure 8: Magnetic field lines form closed loops; hence the net magnetic flux through any closed surface is zero.

Therefore,

$$\boxed{\oint\!\!\!\oint_S \mathbf{B} \cdot d\mathbf{S} = 0}$$

which is the integral form of

$$\boxed{\nabla \cdot \mathbf{B} = 0.}$$

Important remark

There are no isolated magnetic charges (magnetic monopoles) in classical electromagnetics. Hence magnetic field lines always form closed loops.

Ampère's law in magnetostatics

Just as Gauss's law relates the electric flux through a closed surface to enclosed charge, Ampère's law relates the circulation of $\mathbf{B}$ around a closed path to the enclosed current.
In free space,

$$\boxed{\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}}$$

which may be compared with Gauss's law in electrostatics:

$$\boxed{\oiint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}.}$$

Comparison with electrostatics. In electrostatics,

$$\boxed{\nabla \times \mathbf{E} = 0 \qquad \Longleftrightarrow \qquad \oint_C \mathbf{E} \cdot d\boldsymbol{\ell} = 0.}$$

In magnetostatics,

$$\boxed{\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \qquad \Longleftrightarrow \qquad \oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}}$$

in free space.

Key Takeaways

1. Around a straight current-carrying wire, $\mathbf{B}$ is in the $\hat{\boldsymbol{\phi}}$ direction.
2. Magnetic field lines form closed loops, so
$$\oiint_S \mathbf{B} \cdot d\mathbf{S} = 0.$$
3. In free space, Ampère's law is
$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}.$$
4. This plays a role in magnetostatics similar to Gauss's law in electrostatics.

Where will we use these equations? A first important example

Magnetic field of an infinitely long straight wire. Consider an infinitely long straight wire carrying current I along $+\hat{z}$. From symmetry,

- the magnetic field is along the azimuthal direction $\hat{\boldsymbol{\phi}}$,
- the magnitude depends only on the distance s from the wire,
- on a circle of radius s, the magnitude B is constant.

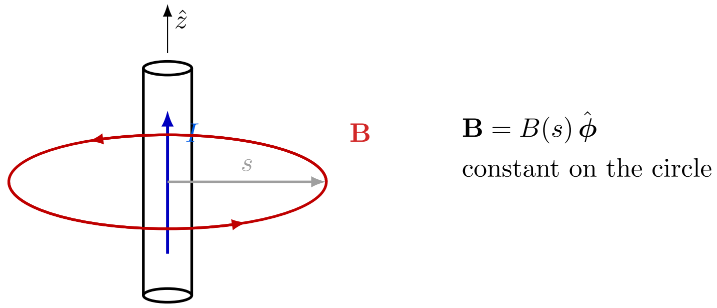


Figure 9: For an infinitely long straight current-carrying wire, magnetic field lines are circular. A circular Amperian loop is the natural symmetry choice.

Using Ampère's law,

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}.$$

Now, on the chosen circular path of radius s,

$$\mathbf{B} \parallel d\boldsymbol{\ell} \qquad \text{and} \qquad |\mathbf{B}| = B(s) = \text{constant}.$$

Hence,

$$B(s) \oint_C d\ell = \mu_0 I$$

that is,

$$B(s)\,(2\pi s) = \mu_0 I.$$

Therefore,

$$\boxed{\mathbf{B}(s) = \hat{\boldsymbol{\phi}}\,\frac{\mu_0 I}{2\pi s}}$$

Remark. The same result can also be obtained by directly integrating the Biot–Savart law. However, Ampère's law makes the derivation much simpler because the symmetry is very clear.

Electrostatic equivalent: infinite line charge

Now compare the above result with the electrostatic field due to an infinitely long line charge of density λ. By symmetry,

- the electric field is in the radial direction $\hat{s}$,
- the magnitude depends only on the distance s from the line charge,
- on a cylindrical Gaussian surface of radius s and length L, the magnitude E is constant on the curved surface.

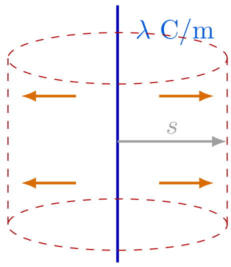


Figure 10: Electrostatic equivalent: an infinite line charge with a cylindrical Gaussian surface. The electric field is radial and constant on the curved surface.

Using Gauss's law,

$$\oint\!\!\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{Q_{\text{enc}}}{\varepsilon_0}.$$

Choose a cylindrical Gaussian surface of radius s and length L. Then the flux through the flat end caps is zero because $\mathbf{E}$ is radial, while the area vectors of the end caps are along $\pm\hat{z}$. So only the curved surface contributes:

$$E(s)\,(2\pi s L) = \frac{\lambda L}{\varepsilon_0}.$$

Thus,

$$\boxed{\mathbf{E}(s) = \hat{s}\,\frac{\lambda}{2\pi\varepsilon_0 s}}$$

Comparison

For the infinite straight wire and the infinite line charge,

$$\boxed{\mathbf{B}(s) = \hat{\phi}\,\frac{\mu_0 I}{2\pi s}} \qquad \text{and} \qquad \boxed{\mathbf{E}(s) = \hat{s}\,\frac{\lambda}{2\pi\varepsilon_0 s}.}$$

Both fields vary as $1/s$, but their directions are different:

- $\mathbf{E}$ is radial,
- $\mathbf{B}$ is azimuthal (circulatory).

Key Takeaways

1. Ampère's law is especially powerful when the magnetic field has a high degree of symmetry.
2. For an infinite straight wire,
$$\mathbf{B}(s) = \hat{\phi}\,\frac{\mu_0 I}{2\pi s}.$$
3. The electrostatic analogue is the field of an infinite line charge:
$$\mathbf{E}(s) = \hat{s}\,\frac{\lambda}{2\pi\varepsilon_0 s}.$$
4. The magnitudes have similar $1/s$ dependence, but the field directions are fundamentally different.

Force between two long parallel current-carrying wires

Consider two infinitely long straight wires carrying currents I_1 and I_2 along $+\hat{z}$, separated by a distance d.

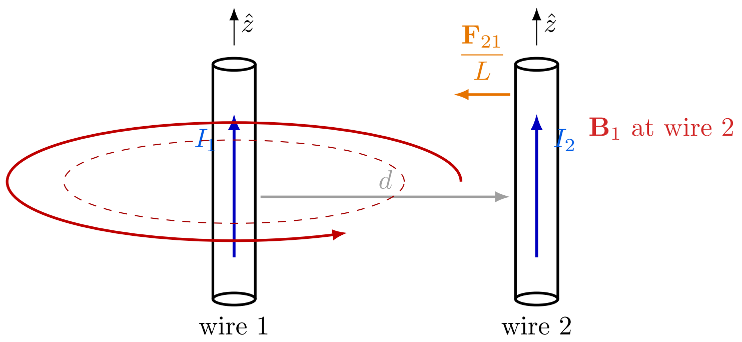


Figure 11: Two long parallel wires carrying currents in the same direction. The magnetic field produced by wire 1 at the location of wire 2 causes an attractive force on wire 2 toward wire 1.

Magnetic field at wire 2 due to wire 1. From the result for an infinitely long straight wire,

$$\boxed{\mathbf{B}_1(d) = \hat{\boldsymbol{\phi}}\,\frac{\mu_0 I_1}{2\pi d}}$$

where the magnitude is

$$\boxed{B_1(d) = \frac{\mu_0 I_1}{2\pi d}.}$$

Force on wire 2 due to wire 1. For a current element,

$$d\mathbf{F} = I\,d\boldsymbol{\ell} \times \mathbf{B}.$$

Hence, for wire 2,

$$\frac{d\mathbf{F}_{21}}{d\ell} = I_2\,\hat{z} \times \mathbf{B}_1.$$

Therefore,

$$\boxed{\frac{d\mathbf{F}_{21}}{d\ell} = I_2\,\hat{z} \times \mathbf{B}_1 = I_2\,\hat{z} \times \left(\hat{\boldsymbol{\phi}}\,\frac{\mu_0 I_1}{2\pi d}\right)}$$

and so the magnitude of force per unit length is

$$\boxed{\frac{F_{21}}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}.}$$

Direction of the force. At the location of wire 2, the direction of $\hat{z} \times \hat{\boldsymbol{\phi}}$ is toward wire 1. Hence wire 2 is attracted toward wire 1.
So,

$$\boxed{\frac{\mathbf{F}_{21}}{L} = -\hat{\mathbf{d}}\,\frac{\mu_0 I_1 I_2}{2\pi d}}$$

where $\hat{\mathbf{d}}$ is the unit vector pointing from wire 1 toward wire 2.

Important conclusion

If two long parallel wires carry currents in the **same direction**, they **attract** each other.

Similarly, if the currents are in opposite directions, the direction of $I_2\,d\boldsymbol{\ell}$ reverses, and hence the force reverses.
Therefore,

$$\boxed{\text{parallel currents in the same direction attract, while opposite currents repel.}}$$

Key Takeaways

1. The magnetic field produced by wire 1 at distance d is
$$B_1 = \frac{\mu_0 I_1}{2\pi d}.$$
2. The force per unit length on wire 2 is
$$\frac{F_{21}}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}.$$
3. Same-direction currents attract; opposite-direction currents repel.

Example: magnetic field at the center of a circular current loop

Consider a circular loop of radius a, carrying current I in the $\hat{\boldsymbol{\phi}}$ direction. We want to find the magnetic field at the center of the loop.

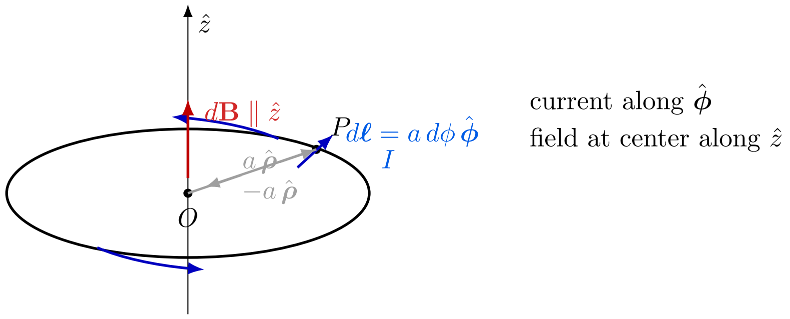


Figure 12: Circular current loop of radius a. For every current element, the contribution $d\mathbf{B}$ at the center is along $\hat{z}$.

Using Biot–Savart law. For a current element,
$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\boldsymbol{\ell} \times \mathbf{R}}{R^3},$$
where $\mathbf{R}$ is the vector from the source point to the observation point.
At the center of the loop,
$$\mathbf{R} = -a\,\hat{\boldsymbol{\rho}}, \qquad R = a, \qquad d\boldsymbol{\ell} = a\,d\phi\,\hat{\boldsymbol{\phi}}.$$
Therefore,
$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,(a\,d\phi\,\hat{\boldsymbol{\phi}}) \times (-a\,\hat{\boldsymbol{\rho}})}{a^3}.$$

Using

$$\hat{\boldsymbol{\phi}} \times (-\hat{\boldsymbol{\rho}}) = \hat{z},$$

we get

$$d\mathbf{B} = \hat{z}\,\frac{\mu_0 I}{4\pi a}\,d\phi.$$

Now integrate over the entire loop:

$$\mathbf{B} = \hat{z}\,\frac{\mu_0 I}{4\pi a}\int_0^{2\pi} d\phi.$$

Hence,

$$\boxed{\mathbf{B} = \hat{z}\,\frac{\mu_0 I}{2a}}$$

Direction of the field

The direction of $\mathbf{B}$ at the center is obtained from the right-hand screw rule. If the current is counter-clockwise when viewed from $+z$, then

$$\mathbf{B} \text{ is along } +\hat{z}.$$

If the current direction is reversed, then $\mathbf{B}$ is along $-\hat{z}$.

Key Takeaways

1. For a circular loop, every current element contributes a magnetic field at the center in the same axial direction.
2. The magnitude of the field at the center is

$$\boxed{B = \frac{\mu_0 I}{2a}}.$$

3. The direction is obtained from the right-hand screw rule.

Why do we focus on a small current-carrying loop?

One may ask: if magnets are already familiar objects, why do we introduce the magnetic field using a *current-carrying loop*?
The reason is that a small current loop is the simplest electrical system that behaves like a magnet. So, by studying a current loop, we can understand the essential physics of magnetic alignment and torque in a clean and quantitative way.

A familiar observation: a magnet aligns with the Earth's magnetic field. A freely suspended magnetic needle tends to rotate and settle along the Earth's magnetic field.

Instead of a magnet, let us study a rectangular current loop. Now place a small rectangular loop carrying current I in a uniform magnetic field $\mathbf{B}$.

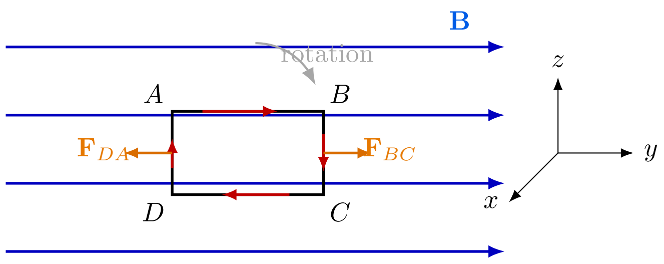


Figure 13: A current loop in a uniform magnetic field. The two vertical sides experience equal and opposite forces, producing a torque.

Force on each side of the loop. The magnetic force on a current element is

$$d\mathbf{F} = I\, d\boldsymbol{\ell} \times \mathbf{B}.$$

For the horizontal sides AB and CD, the current is parallel or anti-parallel to $\mathbf{B}$, so

$$\boxed{\mathbf{F}_{AB} = \mathbf{F}_{CD} = \mathbf{0}.}$$

For the two vertical sides,

$$\mathbf{F}_{BC} = I\, d\, \hat{z} \times \mathbf{B}, \qquad \mathbf{F}_{DA} = I\, d\, (-\hat{z}) \times \mathbf{B},$$

so the two forces are equal in magnitude and opposite in direction.
Hence,

$$\boxed{\mathbf{F}_{\text{net}} = \mathbf{0},}$$

that is, there is *no net translational force* on the loop in a uniform magnetic field.

But the loop still rotates. Why? Although the net force is zero, the two equal and opposite forces act at different locations. Therefore they form a **couple**, which produces a **torque**.
If the width of the loop is b, then the torque magnitude is

$$\boxed{\tau = bF = b\,(IdB) = I\,A\,B}$$

for the orientation shown, where

$$A = bd$$

is the area of the loop.
More generally, if θ is the angle between the loop normal $\hat{n}$ and $\mathbf{B}$, then

$$\boxed{\tau = IAB\sin\theta.}$$

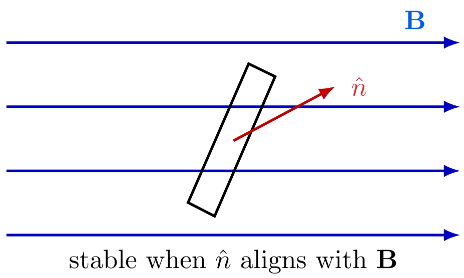


Figure 14: A current loop rotates until its normal aligns with the external magnetic field.

So what happens physically? The loop starts rotating under the action of this torque. As it rotates, the torque changes, and finally the loop settles in the orientation where its normal aligns with the magnetic field.

Physical picture

If there is damping (for example, air friction or internal losses), the loop will not keep oscillating forever. It eventually settles into a stable orientation where its magnetic effect aligns with the external field.

Why this helps us understand magnets

Now comes the key idea.
A magnet can be viewed, in a simplified microscopic picture, as a material containing a huge number of tiny circulating currents (or equivalently tiny magnetic dipoles arising from electron orbital motion and spin). So the behavior of a small current loop is not artificial — it is the simplest model of a magnetic dipole.

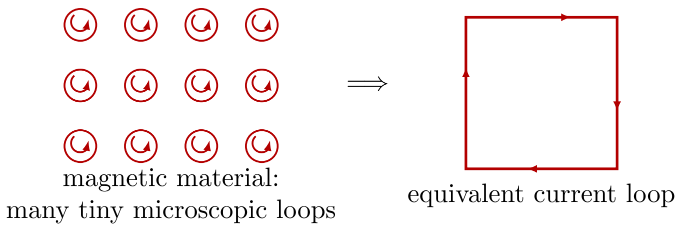


Figure 15: A magnet may be viewed, in a simplified way, as a collection of many tiny current loops or microscopic magnetic dipoles.

Conclusion

A **small current-carrying loop** is the simplest and cleanest model of a **magnet**.
So, by studying the force and torque on a current loop in a magnetic field, we are also learning the basic physics of how magnets behave:

$$\boxed{\text{current loop} \iff \text{magnetic dipole (magnet-like object).}}$$

This is why the current loop becomes a central object in magnetostatics.